Lumbar Spine Degeneration Classification

CS583 Final Project - Mayank Choudhary, Raghav Aggarwal, Nihaar Chhaba, Aneeket Kumar

The Challenge of Diagnosing Chronic Back Pain

Chronic Back Pain Epidemic

Lumbar spine degeneration is a leading cause of persistent back pain, impacting millions globally.

Manual MRI Review Burden

Radiologists face a time-consuming task, manually sifting through hundreds of MRI slices. This process is inherently slow and can lead to subjective interpretations and inconsistencies.

Our Deep Learning Solution

We aim to develop an AI-powered pipeline to classify the severity of degenerative spinal conditions (Normal/Mild, Moderate, Severe), offering a robust tool to assist radiologists and enhance diagnostic accuracy and consistency.

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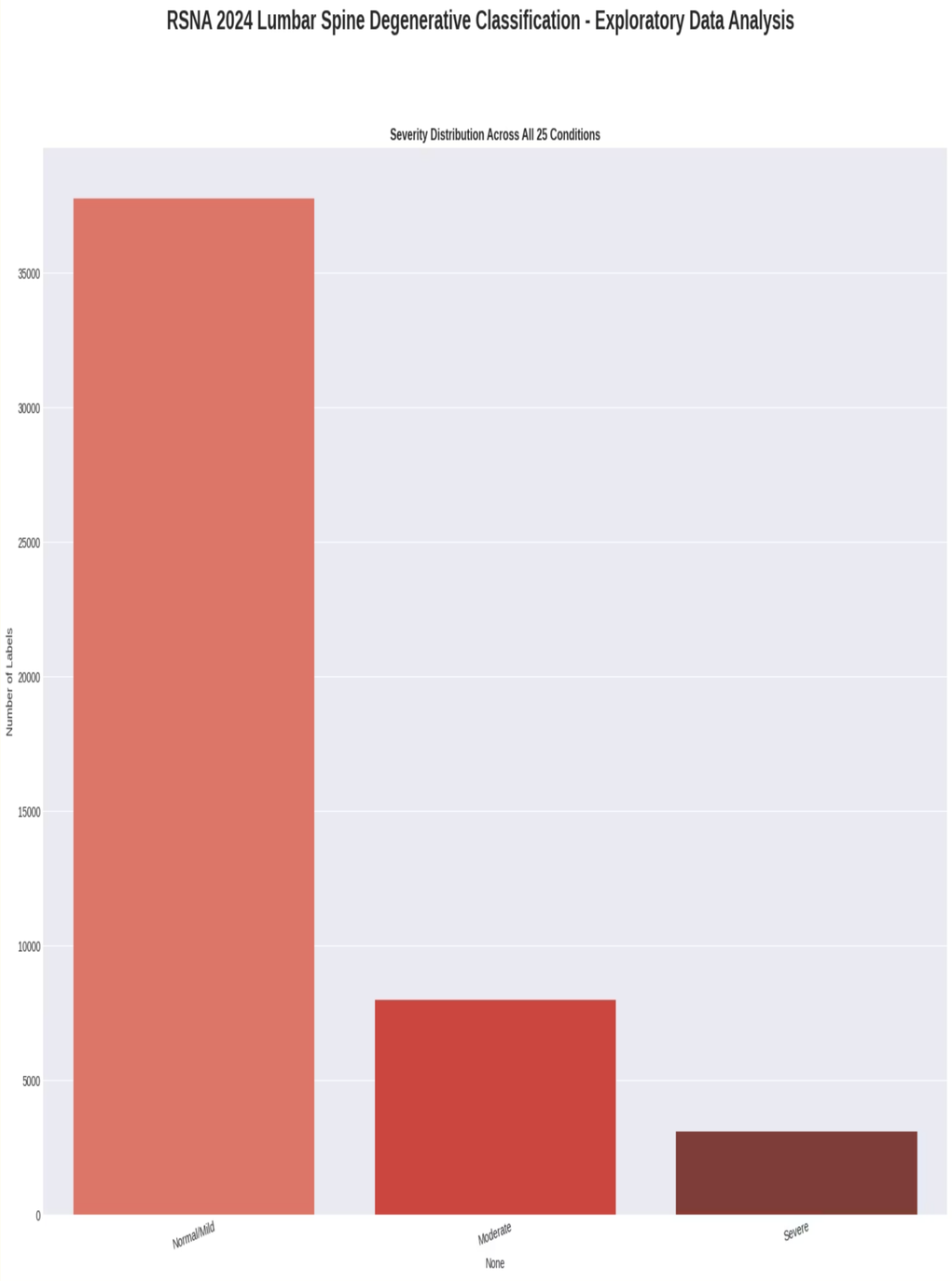

Understanding Our Dataset: RSNA 2024

- The RSNA 2024 Lumbar Spine Degenerative Classification dataset is comprised of approximately 3,000 MRI studies.
- These studies encompass over 18,000 meticulously labeled spinal levels.
- Multiple MRI modalities are included: Sagittal T1, Sagittal T2/STIR, and Axial T2 sequences.
- Each vertebral level is labeled across 25 distinct condition types.
- A significant challenge is the pronounced class imbalance, with Normal/Mild cases outnumbering Severe cases by a ratio of approximately 12.4 : 1.

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Robust Preprocessing for Optimal Model Input

Load DICOM

Utilizing `pydicom` for efficient loading of raw DICOM MRI files.

Slice Selection

Focused selection of Sagittal T2 slices, crucial for spinal degeneration assessment.

Enhancement

Normalization and application of CLAHE (Contrast Limited Adaptive Histogram Equalization) to improve image contrast.

Resizing & Stacking

Images resized to 224x224 or 256x256, followed by stacking 3-5 slices into 2.5D tensors to capture volumetric information.

Dataset Construction

Final construction of the RSYNA Spine Dataset class, ready for model ingestion.

Original MRI Slice

CLAHE Enhanced Slice

Comparison highlighting the improved visibility of anatomical structures after CLAHE enhancement.

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Exploring Diverse Model Architectures

Baseline CNN

A foundational Convolutional Neural Network, serving as our initial benchmark.

EfficientNet-B3

An optimized CNN architecture balancing depth, width, and resolution for efficiency.

ResNet50

A deep residual network, known for mitigating vanishing gradients in very deep models.

Vision Transformer (ViT)

Leveraging self-attention mechanisms to process images as sequences of patches.

SpineModel (Custom CNN)

Our bespoke CNN classifier, tailored specifically for lumbar spine analysis.

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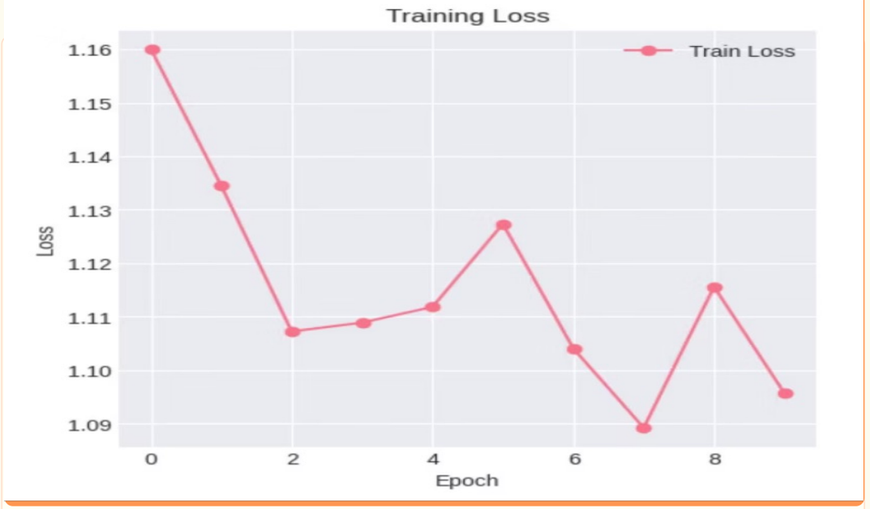
Training Methodology

- All models underwent training for 10 epochs.
- To address severe class imbalance, CrossEntropyLoss with class weights was employed.
- Training was accelerated using dedicated GPU hardware.
- A consistent batch size of 16 was maintained across all experiments.

Training Loss Curve

Illustrates the model's error reduction over epochs.



Training Accuracy Curve

Depicts the model's performance improvement on the training data over time.

Baseline CNN Performance: A Foundation

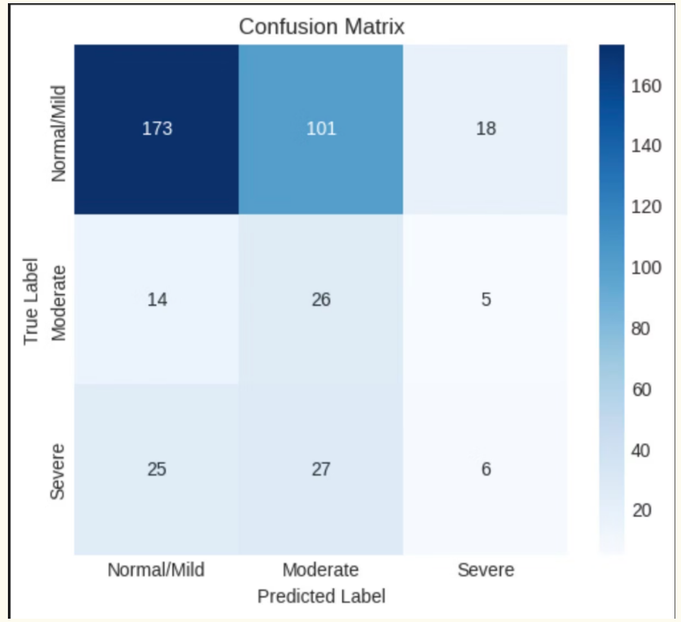
Performance Metrics

- **Accuracy:** 52%
- **Normal/Mild F1 Score:** 0.69
- **Moderate F1 Score:** 0.26
- **Severe F1 Score:** 0.14

> The baseline model showed reasonable performance for the prevalent Normal/Mild class but struggled significantly with accurately identifying Moderate and Severe cases, primarily due to their scarcity in the training data.

Confusion Matrix Summary

The model exhibited strong classification for Normal/Mild cases. However, Severe cases were frequently misclassified as Normal/Mild or Moderate due to their limited representation in the dataset.



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Enhanced Performance: EfficientNet & ResNet

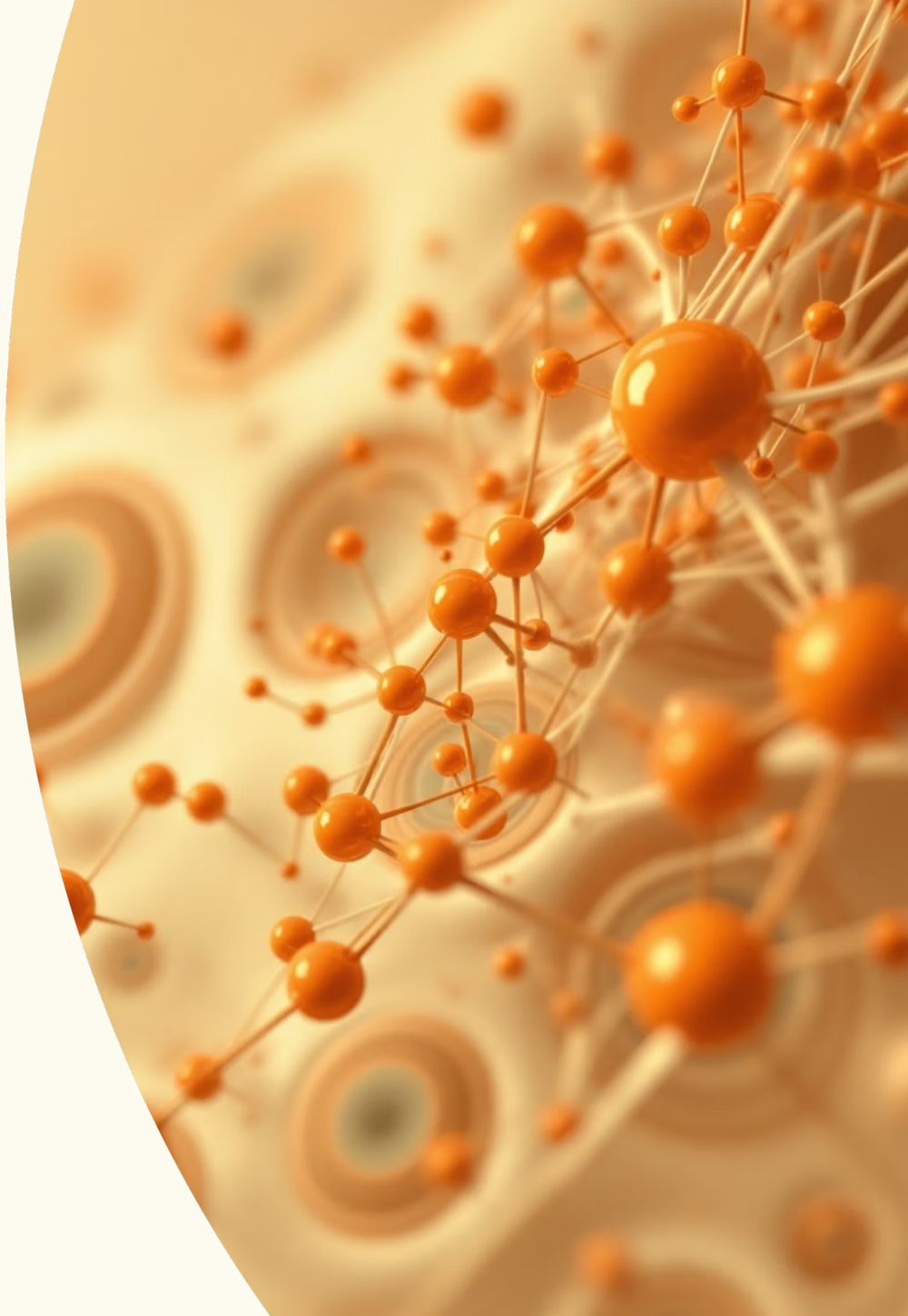
EfficientNet-B3

- **Accuracy:** 65%
- **Normal/Mild F1:** 0.80
- **Moderate F1:** 0.20
- **Severe F1:** 0.30

ResNet50

- **Accuracy:** 62%
- **Normal/Mild F1:** 0.77
- **Moderate F1:** 0.21
- **Severe F1:** 0.28

> These advanced models demonstrated a notable improvement over the baseline, particularly in overall accuracy and the Normal/Mild F1 score. However, they continued to face significant challenges in precisely classifying the less frequent Moderate and Severe degeneration cases, emphasizing the persistent impact of data imbalance.



Vision Transformer and Our Best Model

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Vision Transformer (ViT-Base)

- **Accuracy:** 62%
- **Normal/Mild F1:** 0.78
- **Moderate F1:** 0.18
- **Severe F1:** 0.29

2

Best Model: SpineModel

- **Accuracy:** 64%
- **Normal/Mild F1:** 0.80
- **Moderate F1:** 0.20
- **Severe F1:** 0.35

Our custom SpineModel, a CNN-based architecture, emerged as the top performer. Interestingly, traditional CNNs outperformed the Vision Transformer, suggesting that for this specific dataset with extreme class imbalance and limited severe cases, CNNs were more effective at feature extraction and classification.

How AI Helped (AI as Teammate)

Code Debugging

AI assisted in fixing DICOM handling and slice alignment.

Architecture Design

Helped in choosing optimal model backbones for the task.

Training Stability

Resolved critical BCE vs. CE loss issues for robust training.

Interpretability

Generated confusion matrices and classification reports for insights.

Documentation

Summarized complex results and experiments efficiently.

EDA Assistance

Created severity distribution and imbalance plots for better understanding.

Throughout the project, AI served as a valuable pair-programmer and debugging assistant, significantly accelerating our experimentation and problem-solving processes.

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Conclusion & Future Directions

What Worked

- Successful development of a comprehensive end-to-end deep learning pipeline.
- Consistent and iterative model improvements from baseline to specialized architectures.

Performance Limitations

- Extreme class imbalance, hindering robust minority class detection.
- Inherent variability in MRI slice quality and presentation.
- Limited availability of training samples for critical minority classes.

Future Work

- Precision ROI cropping using anatomical landmarks.
- Exploration of 3D CNNs or transformer-based volumetric models.
- Multi-label learning approach for all 25 conditions.
- Advanced synthetic data augmentation for severe cases.
- Integration of improved attention-based interpretability methods.

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